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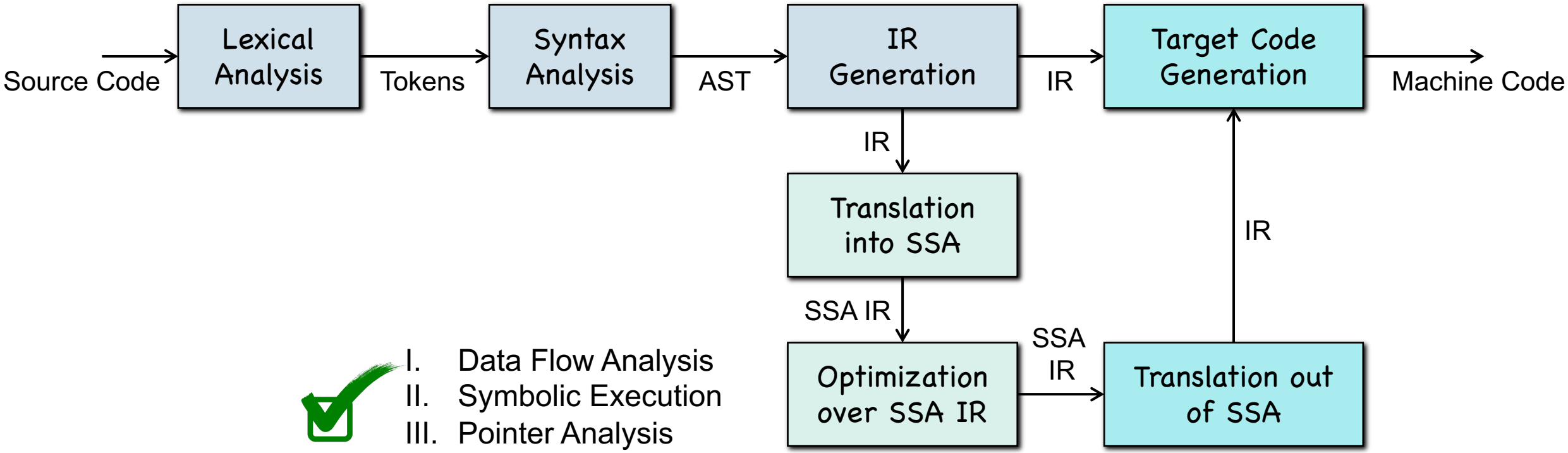
Recap-2
The Compilers' Middle End

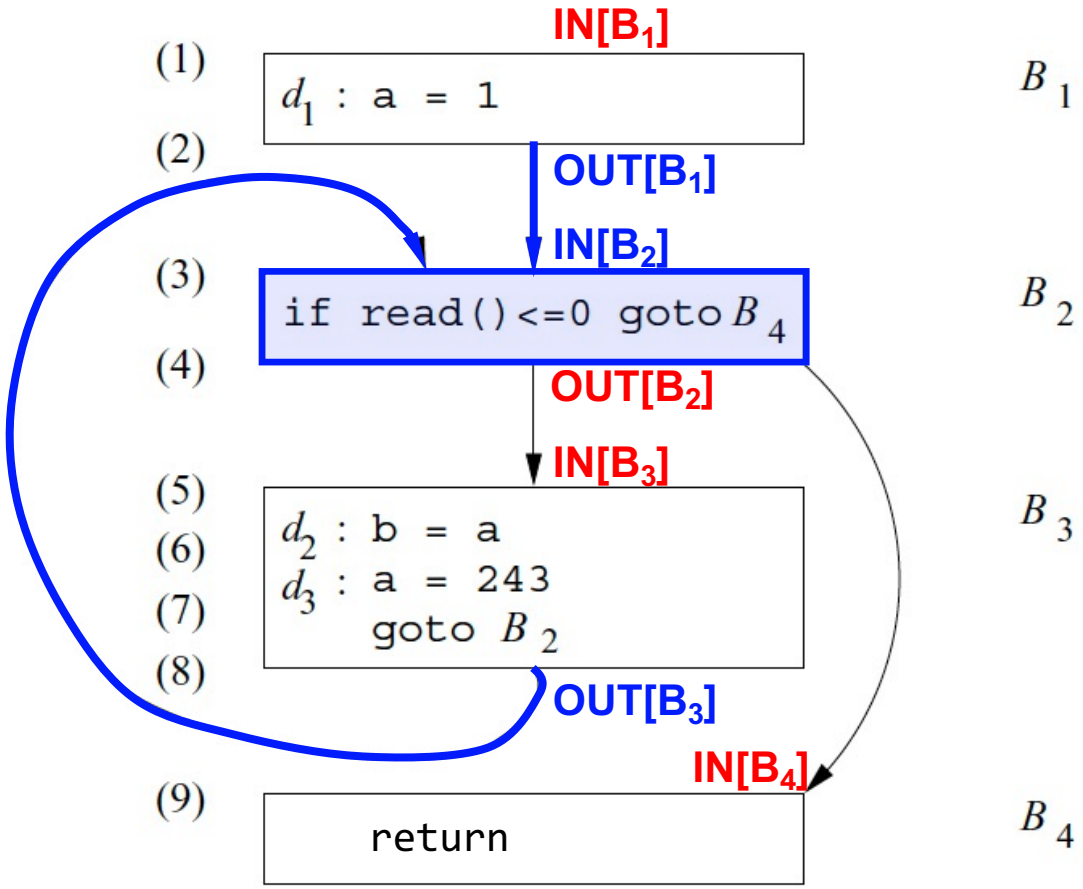
Middle End of Compilers



I. Data Flow Analysis
II. Symbolic Execution
III. Pointer Analysis
IV. Datalog-Based Analysis

PART I: Data Flow Analysis

Data Flow Analysis



- IN[B]/OUT[B]
  - Set of data flow facts before and after a block
- Constraints
  - OUT[B] = f_B(IN[B])
  - IN[B] = $\wedge_{P \in preds(B)}$ OUT[P]

Data Flow Analysis

- To perform data flow analysis, define
  - transfer function f for each statement/block
  - merge function ∧ at the joint point of multiple paths
  - the initial set of data flow facts for each block

- Data flow analysis produces
  - IN[B]/OUT[B] that include data flow facts at a program point
  - The resulting data flow facts are always true during program execution

The Worklist Algorithm

- **ForEach** basic block B: initialize IN[B] and OUT[B]; **EndFor**
- worklist = set of all basic blocks;

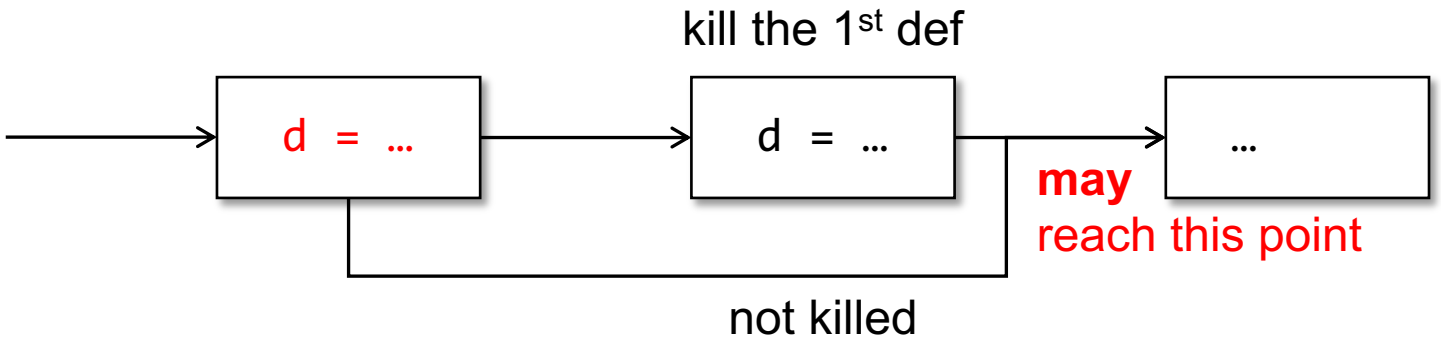
- **While** (!worklist.empty()) **Do**
- B = worklist.pop();
- IN[B] = $\wedge_{P \in preds(B)}$ OUT[P]; OUT[B] = f_B(IN[B])
- **If** OUT[B] changed: worklist.push(all B's successors); **EndIf**
- **EndWhile**

Classic DFA

- Reaching Definition
- Available Expressions
- Live Variables

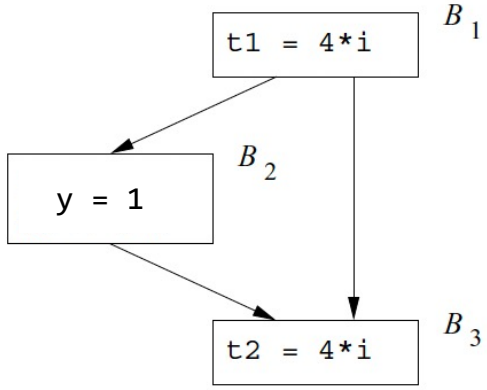
Reaching Definitions

- A definition *d* may reach a program point, if there is a path from *d* to the program point such that *d* is not killed along the path.

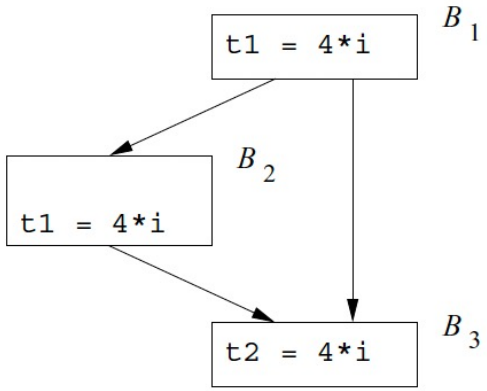


Available Expressions

- An expression x + y is available at a point p if
  - every path from the entry node to p evaluates x + y, and
  - after the last such evaluation prior to reaching p, there are no subsequent assignments to x or y.



4 * i is available before B_3



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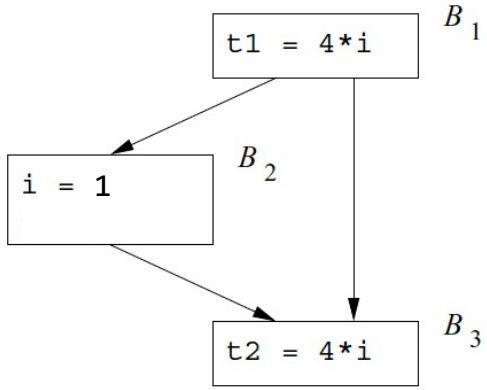


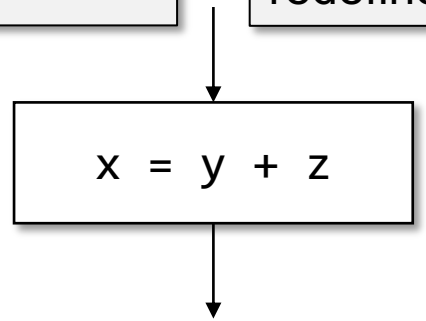
4 * i is **NOT** available before B_3

Live Variable Analysis

- We wish to know if the value of a variable x at point p can be used along the path starting from p.

The values of y, z at this point are **live** as we can use them in the next statement.

The value of x at this point is **dead** because x will be redefined and the old value cannot be used any longer.



Transfer & Merging Functions

- Transfer Functions?
- Merge Functions?
- Forward or Backward Analysis?

PART II: Symbolic Execution

Symbolic Execution

- Path Sensitivity
- Flow Sensitivity
- Context Sensitivity

Symbolic Execution

- Path Explosion
- Constraint Solving
- External Function Call

Solving Path Constraints

- How can we check the satisfiability of a constraint?

$$\alpha_a \neq 0 \wedge \alpha_b = 0 \wedge \neg(2(\alpha_a + \alpha_b) - 4 \neq 0)$$

- Constraint solver
  - Satisfiable => A possible solution
  - Unsatisfiable
  - Unknown (due to timeout)

https://github.com/Z3Prover/z3

https://cvc5.github.io

Solving Path Constraints

- Propositional Logic
- Satisfiability (SAT) --- DPLL and CDCL

- First Order Logic
- Satisfiability Modulo Theories (SMT)
- The Bit Vector Theory

The SAT Problem

- Deciding satisfiability of a formula F in PL

- **Step 1**: Transforming F to CNF by Tseitin Transformation
- **Step 2**: Invoke the DPLL algorithm

What is CNF?
Why not DNF or NNF?

DPLL

- $(p \vee q \vee \neg r) \wedge (p \vee q \vee r) \wedge (p \vee \neg q) \wedge \neg p$

- **Goal**: Find a solution to satisfy the input formula

- Repeat the following steps:
- (1) Choose a variable, p, assign true or false to the variable
- (2) Propagate the value of p

DPLL

- $(p \vee q \vee \neg r) \wedge (p \vee q \vee r) \wedge (p \vee \neg q) \wedge \neg p$

```
bool DPLL (F, I) {
    if (F == false) return UNSAT;
    if (F == true)  return SAT;

    p = choose(F);
    bool ret = DPLL(F[p→true], I[p→true])
    if (ret == SAT) return SAT;

    return DPLL(F[p→false], I[p→ false]);
}
```

DPLL

- $(p \vee q \vee \neg r) \wedge (p \vee q \vee r) \wedge (p \vee \neg q) \wedge \neg p$

```
bool DPLL (F, I) {
    if (F == false) return UNSAT;
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    bool ret = DPLL(F[p→true], I[p→true])
    if (ret == SAT) retu

    return DPLL(F[p→fal
}
```

Optimizations may be applied!

Theorem [Resolution]

$$(a_0 \vee a_1 \vee \cdots \vee a_n \vee \boldsymbol{c}) \wedge (b_0 \vee b_1 \vee \cdots \vee b_m \vee \neg \boldsymbol{c})$$

can be simplified into

$$(a_0 \vee a_1 \vee \cdots \vee a_n \vee b_0 \vee b_1 \vee \cdots \vee b_m)$$

DPLL

- $(p \vee q \vee \boxed{\neg r}) \wedge (p \vee q \vee \boxed{r}) \wedge (p \vee \neg q) \wedge \neg p$
- $(p \vee \boxed{q}) \wedge (p \vee \boxed{\neg q}) \wedge \neg p$
- $p \wedge \neg p$ **False! Unsatisfiable!**

Theorem [Resolution]

$$(a_0 \vee a_1 \vee \cdots \vee a_n \vee \boldsymbol{c}) \wedge (b_0 \vee b_1 \vee \cdots \vee b_m \vee \neg \boldsymbol{c})$$

can be simplified into

$$(a_0 \vee a_1 \vee \cdots \vee a_n \vee b_0 \vee b_1 \vee \cdots \vee b_m)$$

Tseitin Transformation

- $(r \Rightarrow p) \Rightarrow (\neg(q \wedge r) \Rightarrow p)$

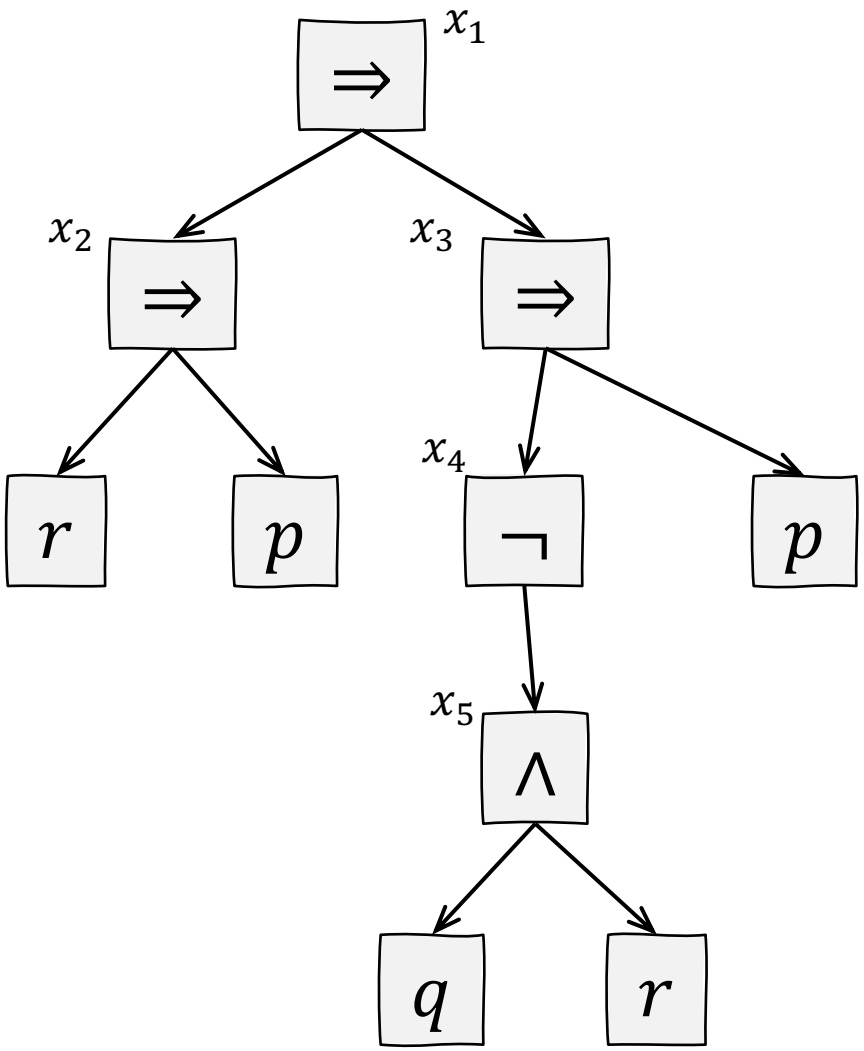
x_2: $(r \Rightarrow p)$

x_5: $(q \wedge r)$

x_4: $\neg(q \wedge r)$

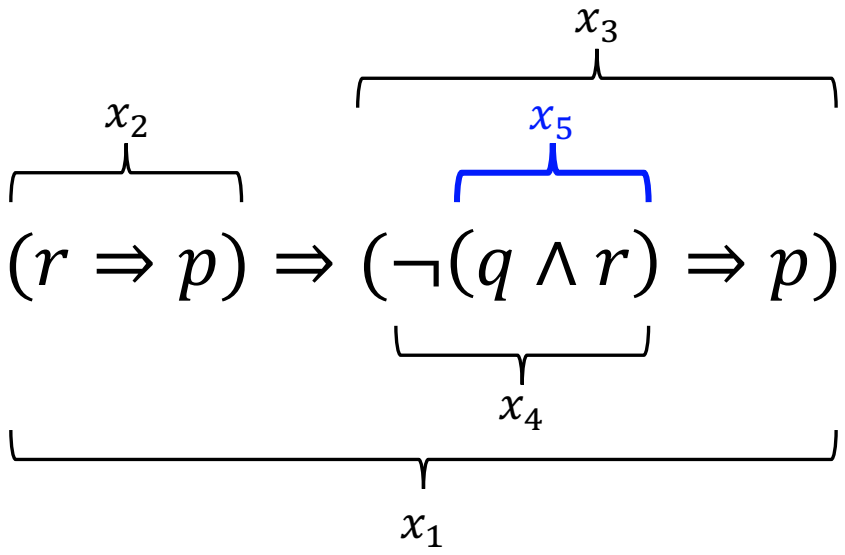
x_3: $(\neg(q \wedge r) \Rightarrow p)$

x_1: $(r \Rightarrow p) \Rightarrow (\neg(q \wedge r) \Rightarrow p)$



Tseitin Transformation

- $(r \Rightarrow p) \Rightarrow (\neg(q \wedge r) \Rightarrow p)$



$$x_1 \wedge (x_1 \Leftrightarrow (x_2 \Rightarrow x_3))$$
$$\wedge\ x_2 \Leftrightarrow (r \Rightarrow p)$$
$$\wedge\ x_3 \Leftrightarrow (x_4 \Rightarrow p)$$
$$\wedge\ x_4 \Leftrightarrow \neg x_5$$
$$\wedge\ x_5 \Leftrightarrow q \wedge r$$

$$x_4 \Rightarrow \neg x_5 \wedge \neg x_5 \Rightarrow x_4$$
$$(\neg x_4 \vee \neg x_5) \wedge (x_5 \vee x_4)$$

Satisfiability Modulo Theories

- For the theory of bit vectors
  - **Step 1**: Word-Level Preprocessing
  - **Step 2**: Bit-Blasting (From FOL to PL)
  - **Step 3**: DPLL/CDCL

Example of Step 2:

$x = [a_7, a_6, \dots, a_0], y = [b_7, b_6, \dots, b_0]$ and $z = [c_7, c_6, \dots, c_0]$ are three 8bit bit vectors

$$x \mid y = z \xrightarrow{\text{Step 2}} \bigwedge_{i=0}^{7} ((a_i \vee b_i) \Leftrightarrow c_i)$$

PART III: Pointer Analysis

Andersen & Steensgaard Algorithm

Inclusion-based pointer analysis (Andersen's Algorithm)

Statement	Constraint	Shorthand
y = &x	pts(y) ⊇ {x}	
y = x	pts(y) ⊇ pts(x)	
*y = x	∀ v ∈ pts(y): pts(v) ⊇ pts(x)	pts(*y) ⊇ pts(x)
y = *x	∀ v ∈ pts(x): pts(y) ⊇ pts(v)	pts(y) ⊇ pts(*x)

Unification-based pointer analysis (Steensgaard's Algorithm)

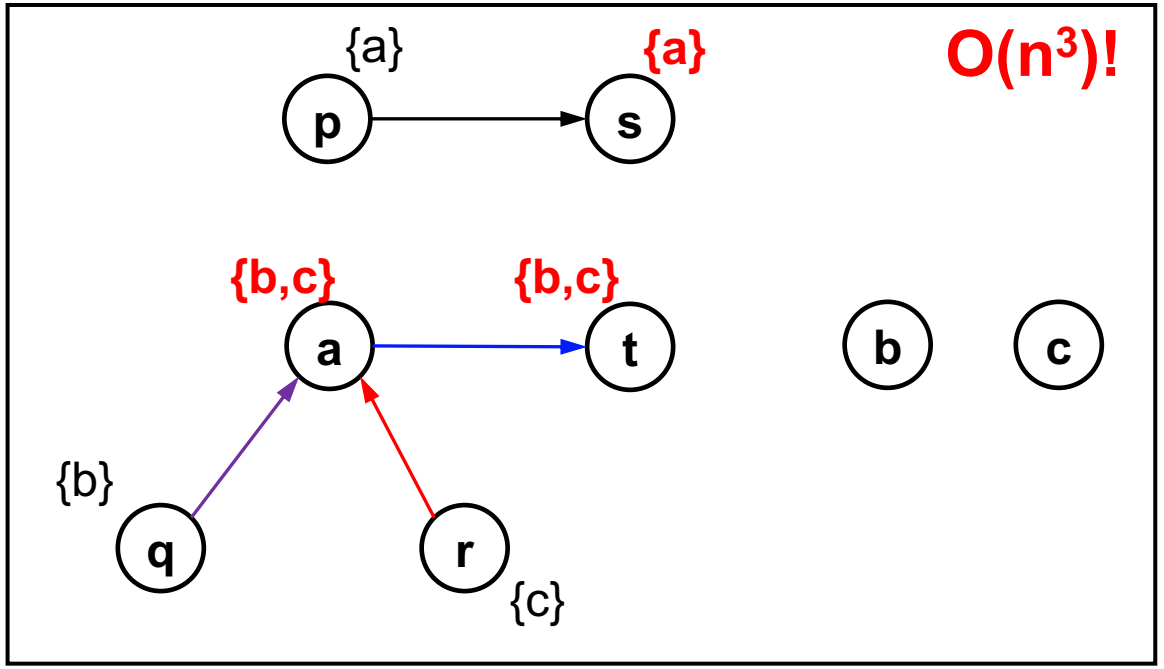
Statement	Constraint	Shorthand
y = &x	pts(y) ⊇ {x}	
y = x	pts(y) = pts(x)	
*y = x	∀ v ∈ pts(y): pts(v) = pts(x)	pts(*y) = pts(x)
y = *x	∀ v ∈ pts(x): pts(y) = pts(v)	pts(y) = pts(*x)

Andersen's Alg. as Graph Closure

- Each graph node x denotes the points-to set $pts(x)$
- Solving the set constraints via a dynamic transitive closure

```
p  =  &a;
q  =  &b;
*p =  q;
r  =  &c;
s  =  p;
t  =  *p;
*s =  r;
```

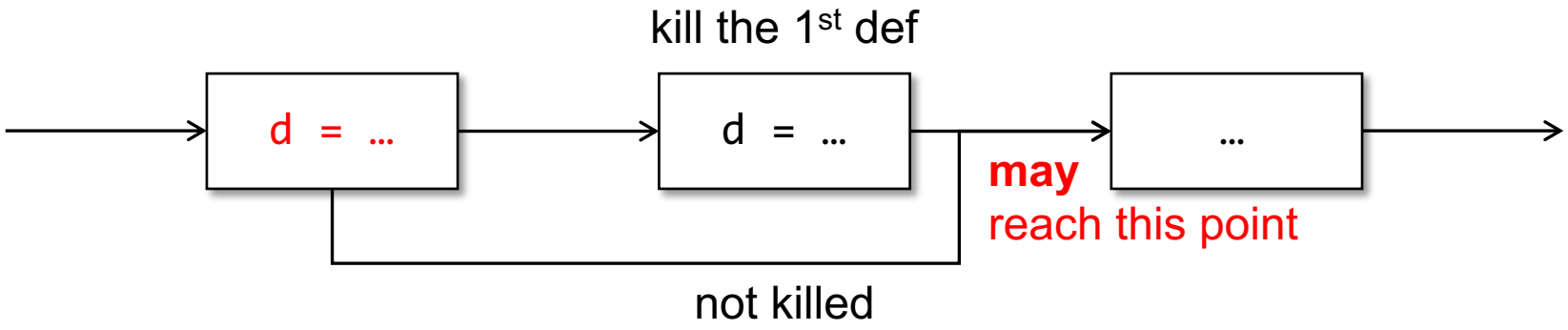
```
...
...
pts(*p) ⊇ pts(q)
...
...
pts(t) ⊇ pts(*p)
pts(*s) ⊇ pts(r)
```



PART IV: Datalog-Based Analysis

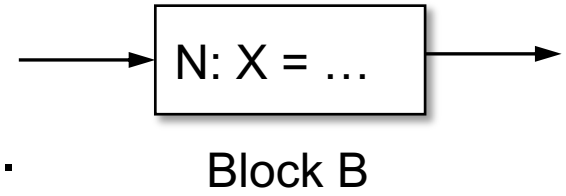
Recap: Reaching Definition

- A definition *d* may reach a program point, if there is a path from *d* to the program point such that *d* is not killed along the path.



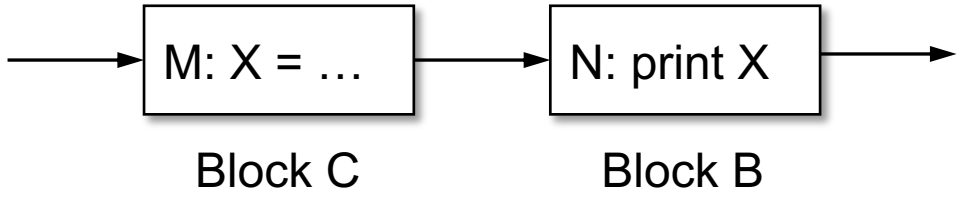
Reaching Definitions by Datalog

- def(B, N, X)
  - the Nth statement in Block B may define the variable X.



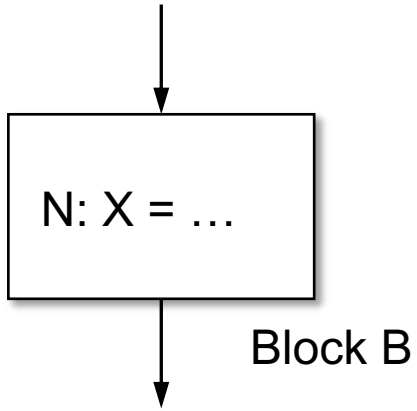
- succ(B, N, C)
  - block C is a successor of block B, and B has N statements.

- rd(B, N, C, M, X)
  - the definition of variable X at the Mth statement of block C reaches the Nth statement in B.



Reaching Definitions by Datalog

- rd(B, N, B, N, X) :- def(B, N, X)
- rd(B, N, C, M, X) :- rd(B, N-1, C, M, X), def(B, N, Y), X≠Y
- rd(B, 0, C, M, X) :- rd(D, N, C, M, X), succ(D, N, B)



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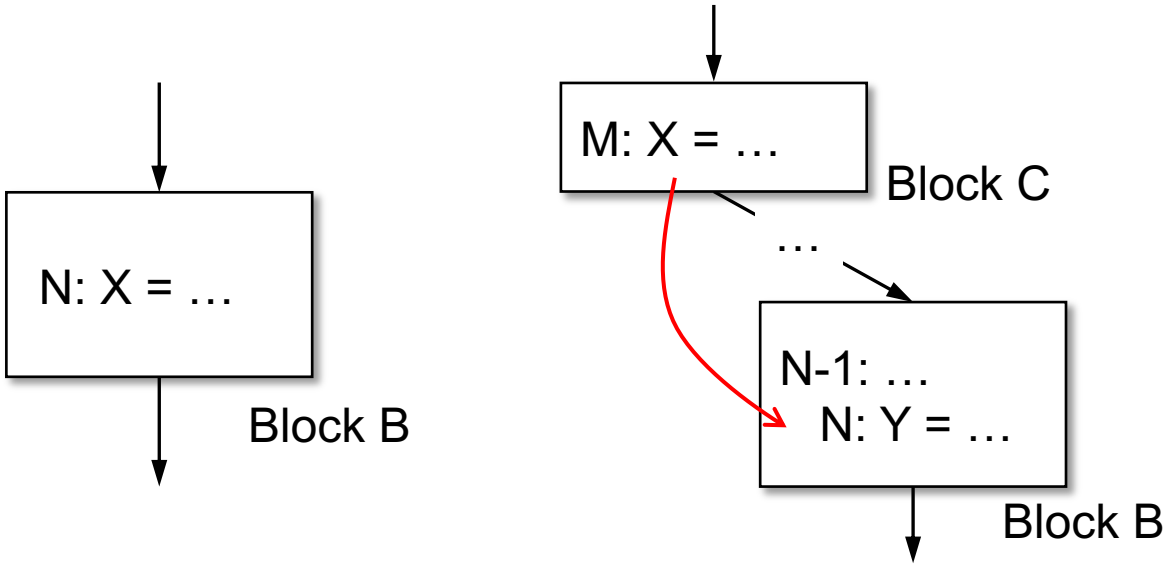

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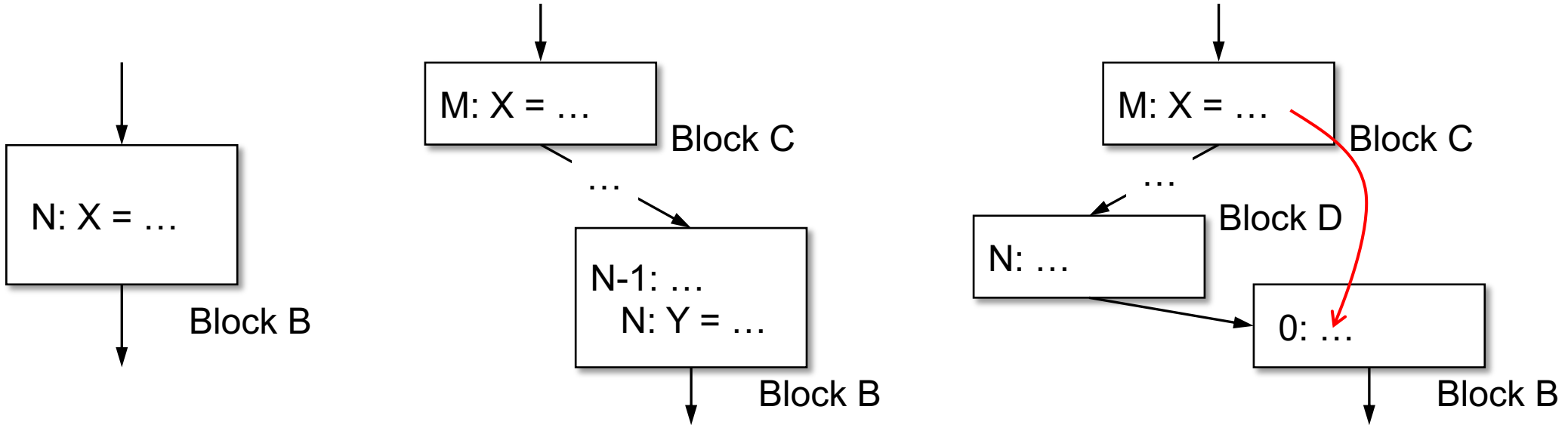

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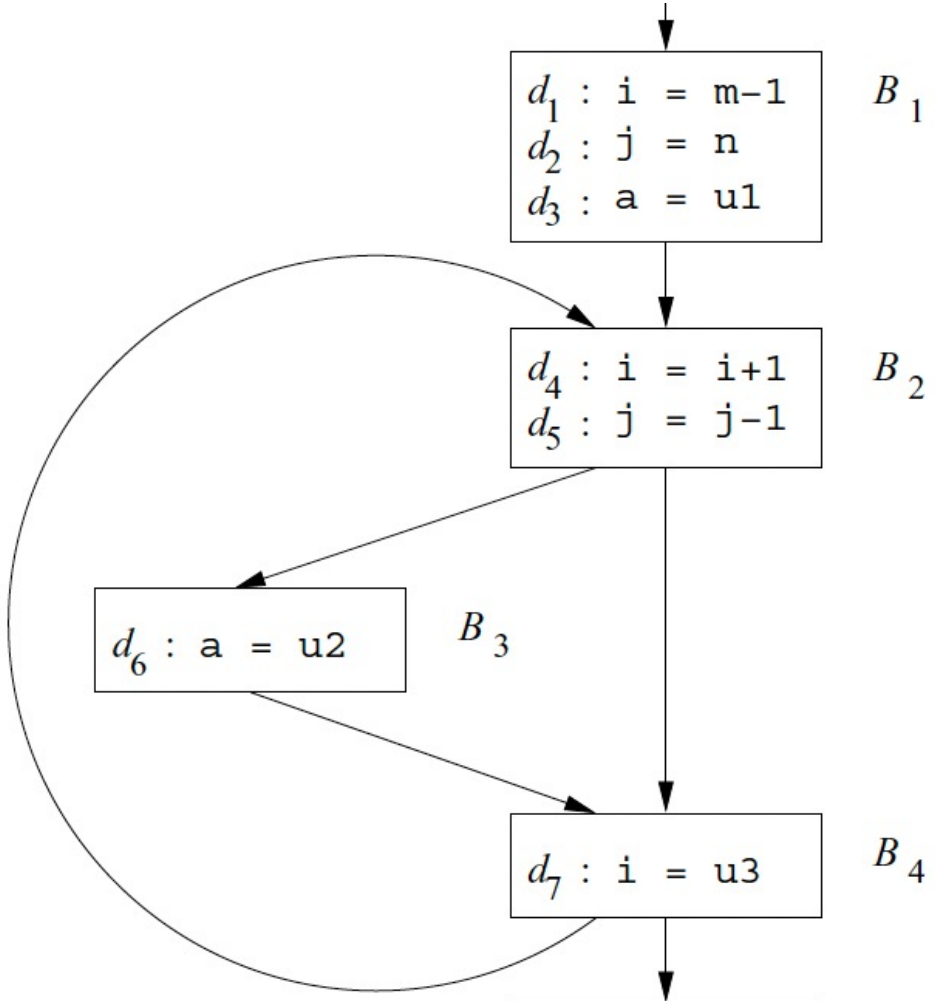

- def(B_1, 1, i)
- def(B_1, 2, j)
- def(B_1, 3, a)
- def(B_2, 1, i)
- def(B_2, 2, j)
- def(B_3, 1, a)
- def(B_4, 1, i)

- succ(B_1, 3, B_2)
- succ(B_2, 2, B_3)
- succ(B_2, 2, B_4)
- succ(B_3, 1, B_4)
- succ(B_4, 1, B_2)

Reaching Definitions by Datalog

- rd(B, N, B, N, X) :- def(B, N, X)
- rd(B, N, C, M, X) :- rd(B, N-1, C, M, X), def(B, N, Y), X≠Y
- rd(B, 0, C, M, X) :- rd(D, N, C, M, X), succ(D, N, B)

- def(B_1, 1, i)
- def(B_1, 2, j)
- def(B_1, 3, a)
- def(B_2, 1, i)
- def(B_2, 2, j)
- def(B_3, 1, a)
- def(B_4, 1, i)
- succ(B_1, 3, B_2)
- succ(B_2, 2, B_3)
- succ(B_2, 2, B_4)
- succ(B_3, 1, B_4)
- succ(B_4, 1, B_2)

Reaching Definitions by Datalog

We just define facts and rules.
Analysis is automatically done by Datalog engines!

Query Example: rd(B_4, 1, B_1, 1, i)